SUPERMARKET SALES

Introduction to Statistics

Bachelor Degree in Artificial Intelligence

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1. MOTIVATION OF THE WORK

This report describes the data selection, preprocessing, and analysis carried out as part of our project. The first main goal was to identify a database that met certain specific criteria, preprocess it so we could prepare it for the subsequent analysis, and perform univariate and bivariate descriptive analysis about the data.

After these objectives were done, the second part was about initially taking a sample, analyzing it, creating a linear regression model, and finally predicting data with it.

We started our research by looking for a reliable website where we could look for a database that met our requirements. We finally found one that fit our requirements and had a large number of variables that would be interesting to work with.

The selected database focuses on supermarket sales data from three different cities in China. It includes variables such as product type, total quantity purchased, customer gender, payment method and time of the month, among other data.

This report will detail each stage of our process, from the initial database selection to the studies that were carried out. As well as delivering a clear overview of our work.

2. DATA SOURCE PRESENTATION

The selected database delves into the realm of supermarket sales, capturing transactional data from three distinct cities across Myanmar. Encompassing variables ranging from product types and quantities purchased to customer gender, payment modalities, and temporal factors, the dataset offers a holistic view of consumer behavior within the supermarket landscape.

This report meticulously documents each phase of our project, traversing from the meticulous database selection process to the intricate analyses conducted thereafter. By offering an exposition of our methodologies and findings, this report aims to provide stakeholders with a comprehensive understanding of our project's scope and outcomes.

3. DATA STRUCTURE AND METADATA

The rows of the data matrix in this dataset represent individual transactions at a supermarket. Each row contains detailed information about a single transaction, capturing various attributes such as the date and time of the purchase, the type of product bought, the branch of the supermarket where the transaction occurred, and customer demographics. Additionally, financial details such as unit price, quantity purchased, tax amount, and total cost are included. These rows provide a comprehensive view of each sale, allowing for in-depth analysis of purchasing patterns and trends.

Here is the metadata table

Column Name	**Description**	**Data Type**	**Example**
Invoice ID	Unique identifier for each transaction	String	750-67-8428
Branch	Branch where the transaction occurred	String	A, B, C
City	City where the branch is located	String	Yangon, Mandalay, Naypyidaw
Customer Type	Type of customer (Member or Normal)	String	Member
Gender	Gender of the customer	String	Female
Product Line	Category of the product purchased	String	Health and beauty
Unit Price	Price per unit of the product	Float	74.69
Quantity	Quantity of the product purchased	Integer	7
Tax 5%	Tax amount (5% of total price)	Float	26.1415
Total	Total amount (including tax)	Float	548.9715
Date	Date of the transaction	Date	01-05-2019

Time	Time of the transaction	Time	13:08
Payment	Payment method used	String	Credit card
cogs	Cost of goods sold	Float	522.83
gross margin percentage	Gross margin percentage per transaction	Float	4.761904762
gross income	Gross income per transaction	Float	26.1415
Rating	Customer rating of the transaction	Float	9.1

The final scope of this study focuses on analyzing customer purchasing behavior and financial performance across different branches of the supermarket. To ensure the integrity and relevance of the analysis, specific inclusion and exclusion criteria have been established for both rows and columns.

Inclusion Criteria:

Rows: Transactions that are complete and contain no missing data in essential fields such as Invoice ID, Branch, City, Product Line, Total, Date, and Payment. In our case, we had no problems with missing data as our database was complete.

Columns: Only columns that provide direct insight into purchasing patterns and financial metrics are included. This encompasses Branch, City, Customer Type, Gender, Product Line, Unit Price, Quantity, Total, Date, Payment, and Rating.

Exclusion Criteria:

Rows: Transactions with any missing or null values in critical fields. Transactions that fall outside the normal operational hours of the supermarkets to avoid anomalies.

Columns: Columns such as cogs, gross margin percentage, and gross income may be excluded if the focus is purely on sales data and customer demographics without delving into detailed financial ratios and margins. Additionally, columns with redundant information or those derived from other columns (e.g., Tax 5% when Total already includes tax) may be excluded to streamline the analysis.

By adhering to these criteria, the study aims to provide a clear and concise analysis of supermarket sales data, highlighting key trends and insights that can inform business decisions and strategy development.

4. PREPROCESSING AND DATA PREPARATION

Although the original database seemed very complete, there were still some changes we had to do to it before we could actually run this data through our Rscripts.

An important step in our data preprocessing was formatting the date variable so it was DD/MM/YYYY and not MM/DD/YYYY. We also had to replace periods with commas. This was done to ensure consistency in the data format and to facilitate further analysis.

Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%	Total	Date	Time	Payment	cogs
834-45-5519	B	Mandalay	Normal	Female	ic accessories	43.00	4	45.451	180.6	3/30/2019	20:48:00	Ewallet	172
593-65-1552	C	Naypyitaw	Normal	Female	ne and lifestyle	69.81	4	13.962	293,202	3/30/2019	20:50:00	Credit card	279.24
227-07-4446	C	Naypyitaw	Member	Female	ic accessories	78.13	10	39.065	820,365	3/30/2019	20:51:00	Cash	781.3
431-66-2305	B	Mandalay	Normal	Female	ic accessories	88.25	9	397.125	8,339,625	3/30/2019	20:51:00	Credit card	794.25
566-19-5475	B	Mandalay	Normal	Male	on accessories	47.97	7	167.895	3,525,795	3/30/2019	20:52:00	Cash	335.79
628-34-3388	C	Naypyitaw	Normal	Male	on accessories	27.38	6	8.214	172,494	3/30/2019	20:54:00	Credit card	164.28
726-29-6793	A	Yangon	Member	Male	ic accessories	24.18	8	9.672	203,112	3/30/2019	20:54:00	Ewallet	193.44
571-94-0759	B	Mandalay	Member	Female	and beverages	74.60	10	37.300	783.3	3/30/2019	20:55:00	Cash	746
715-20-1673	B	Mandalay	Normal	Male	ic accessories	28.38	5	7.095	148,995	3/30/2019	20:57:00	Cash	141.9
700-81-1757	A	Yangon	Normal	Female	ic accessories	26.31	5	1,415,468	1,381,275	3/30/2019	20:59:00	Credit card	131.55
235-06-8510	C	Naypyitaw	Member	Male	ne and lifestyle	85.72	3	12.858	270,018	3/30/2019	20:59:00	Ewallet	257.16
418-05-0656	B	Mandalay	Normal	Female	on accessories	25.56	7	8.946	187,866	3/29/2019	20:42:00	Cash	178.92
190-14-3147	B	Mandalay	Normal	Female	alth and beauty	17.97	4	3.594	75,474	3/29/2019	20:43:00	Ewallet	71.88
151-33-7434	B	Mandalay	Normal	Female	and beverages	67.77	1	725.068	711,585	3/29/2019	20:43:00	Credit card	67.77
314-23-4520	C	Naypyitaw	Member	Male	alth and beauty	81.23	7	284.305	5,970,405	3/29/2019	20:44:00	Cash	568.61
521-18-7827	C	Naypyitaw	Member	Male	ne and lifestyle	39.39	5	2,401,715	2,067,975	3/29/2019	20:46:00	Credit card	196.95
745-74-0715	A	Yangon	Normal	Male	ic accessories	58.03	2	5.803	121,863	3/29/2019	20:46:00	Ewallet	116.06
556-72-8512	C	Naypyitaw	Normal	Male	ne and lifestyle	22.96	1	1.148	24,108	3/29/2019	20:47:00	Cash	22.96
642-32-2990	A	Yangon	Normal	Female	and beverages	10.96	10	5.480	115.08	3/29/2019	20:48:00	Ewallet	109.6
373-09-4567	C	Naypyitaw	Normal	Male	and beverages	77.56	10	38.780	814.38	3/28/2019	20:35:00	Ewallet	775.6
120-06-4233	C	Naypyitaw	Normal	Male	ic accessories	30.61	6	9.183	192,843	3/28/2019	20:36:00	Cash	183.66
726-27-2396	A	Yangon	Normal	Female	alth and beauty	77.50	5	19.375	406,875	3/28/2019	20:36:00	Ewallet	387.5
471-41-2823	C	Naypyitaw	Normal	Male	and beverages	99.79	2	9.979	209,559	3/28/2019	20:37:00	Ewallet	199.58
760-27-5490	C	Naypyitaw	Normal	Male	on accessories	15.62	8	6.248	131,208	3/28/2019	20:37:00	Ewallet	124.96
364-34-2972	C	Naypyitaw	Member	Male	ic accessories	96.82	3	14.523	304,983	3/28/2019	20:37:00	Cash	290.46
594-34-4444	A	Yangon	Normal	Male	ic accessories	97.16	1	4.858	102,018	3/28/2019	20:38:00	Ewallet	97.16
378-24-2715	B	Mandalay	Normal	Male	ne and lifestyle	53.44	2	5.344	112,224	3/28/2019	20:38:00	Ewallet	106.88

Part of the database before having been preprocessed. Note the American date format and the use of dots to mark decimal places.

Additionally, we created two new categorical variables: Time of month and Time of day. These two variables would prove to be useful later on when making bivariate graphs.

Tax 5%	Total	Date	Time	Payment	cogs	gross margin percentage	gross income	Rating	Time of the month	Time of the day
21,783	457,443	01/01/2019	10:39	Credit card	435,66	4,761904762	21,783	6,9	Beginning	Morning
42,315	888,615	01/01/2019	11:36	Credit card	846,3	4,761904762	42,315	9	Beginning	Morning
8,766	184,086	01/01/2019	11:40	Ewallet	175,32	4,761904762	8,766	5	Beginning	Morning
6,287	132,027	01/01/2019	11:43	Cash	125,74	4,761904762	6,287	5	Beginning	Morning
29,583	621,243	01/01/2019	13:55	Cash	591,66	4,761904762	29,583	7,7	Beginning	Midday
37,11	779,31	01/01/2019	14:42	Credit card	742,2	4,761904762	37,11	4,3	Beginning	Midday
19,036	399,756	01/01/2019	14:47	Cash	380,72	4,761904762	19,036	5,7	Beginning	Midday
6,322	132,762	01/01/2019	15:51	Cash	126,44	4,761904762	6,322	8,5	Beginning	Afternoon
22,413	470,673	01/01/2019	19:07	Cash	448,26	4,761904762	22,413	6,7	Beginning	Evening
8,448	177,408	01/01/2019	19:31	Cash	168,96	4,761904762	8,448	6,3	Beginning	Evening
18,49	388,29	01/01/2019	19:48	Credit card	369,8	4,761904762	18,49	7	Beginning	Evening
5,408	113,568	01/01/2019	20:26	Ewallet	108,16	4,761904762	5,408	6,9	Beginning	Evening
17,105	359,205	02/01/2019	13:00	Cash	342,1	4,761904762	17,105	5,1	Beginning	Midday
18,2745	383,7645	02/01/2019	13:40	Cash	365,49	4,761904762	18,2745	7	Beginning	Midday
12,498	262,458	02/01/2019	15:24	Ewallet	249,96	4,761904762	12,498	5,6	Beginning	Afternoon
12,668	266,028	02/01/2019	16:19	Credit card	253,36	4,761904762	12,668	5,6	Beginning	Afternoon
2,1235	44,5935	02/01/2019	16:57	Cash	42,47	4,761904762	2,1235	5,7	Beginning	Afternoon
9,982	209,622	02/01/2019	18:09	Credit card	199,64	4,761904762	9,982	6,7	Beginning	Afternoon
6,603	138,663	02/01/2019	18:50	Cash	132,06	4,761904762	6,603	7,6	Beginning	Afternoon
13,389	281,169	02/01/2019	20:08	Credit card	267,78	4,761904762	13,389	5,1	Beginning	Evening
3,794	79,674	03/01/2019	10:30	Credit card	75,88	4,761904762	3,794	7,1	Beginning	Morning
17,5025	367,5525	03/01/2019	11:36	Ewallet	350,05	4,761904762	17,5025	5,5	Beginning	Morning
10,3635	217,6335	03/01/2019	12:02	Ewallet	207,27	4,761904762	10,3635	7,9	Beginning	Midday
4,669	98,049	03/01/2019	13:07	Cash	93,38	4,761904762	4,669	9,6	Beginning	Midday
16,7725	352,2225	03/01/2019	16:47	Credit card	335,45	4,761904762	16,7725	9,1	Beginning	Afternoon
24,8	520,8	03/01/2019	19:08	Credit card	496	4,761904762	24,8	6,2	Beginning	Evening

Section of the preprocessed database. Note the addition of 2 new variables: Time of month and Time of day

Once we had finished with the preprocessing, the database was ready for all of our needed uses, such as the creation of univariate and bivariate graphs, data inference and hypothesis testing.

5. STATISTICAL DESCRIPTIVE ANALYSIS

a. UNIVARIATE

We made graphs for each variable, consisting of box plots, bar plots, pie charts and histograms. The tool that we've used to make these graphs is R studio with a piece of code that the professor provided to us and that we had then edited to make it suitable for our database.

The graphs that we've done are the following ones, dividing them depending on the type of variable that it is.

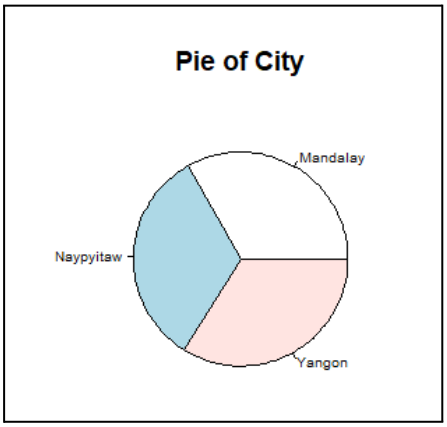
Categorical variables:

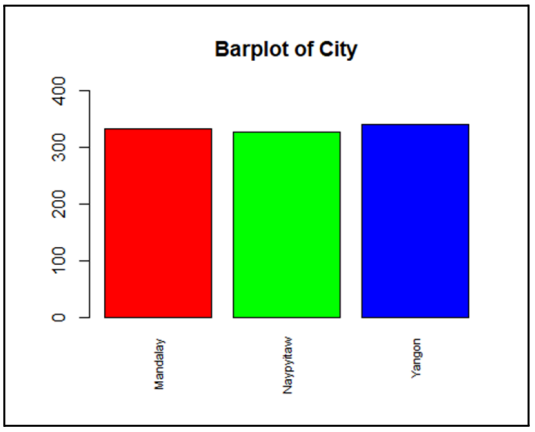
In the categorical variables we can find nominal, binary and ordinal variables too.

- Nominal variables:

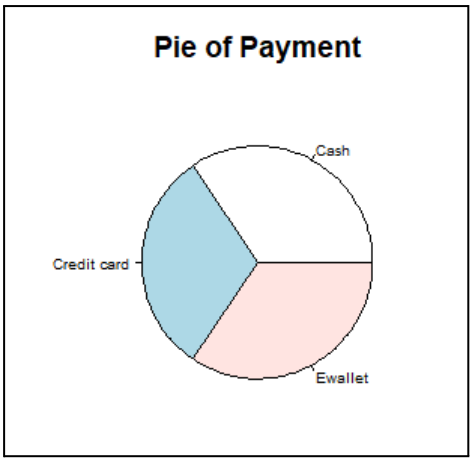
  In our database all the nominal variables have a pie chart and a bar plot and they are:

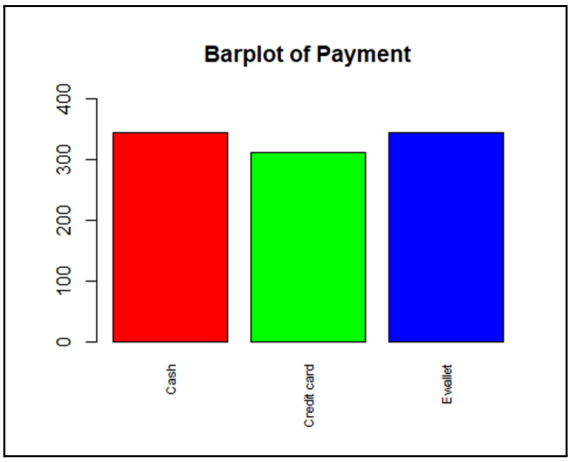
  - The type of city, of which we have a pie chart and a bar plot. Here we can see that the distribution is very uniform, as each city has almost the same number of customers.



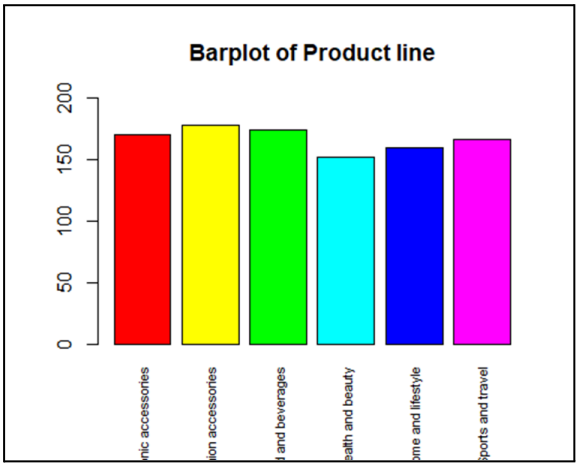


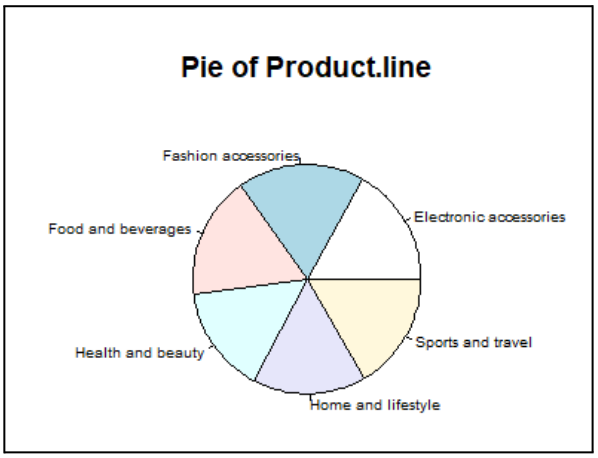
  - The type of payment. Here we can see that the distribution is quite uniform too, even though the payment with credit card is less common than the others.



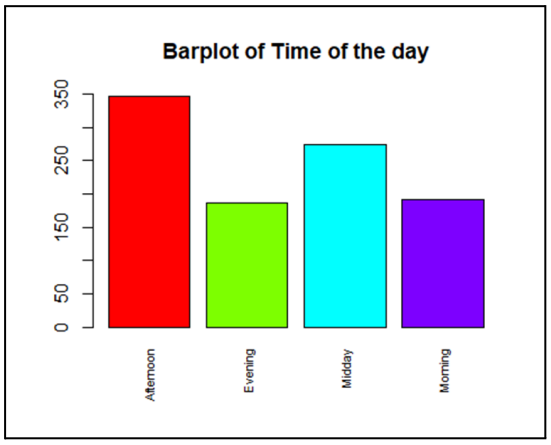


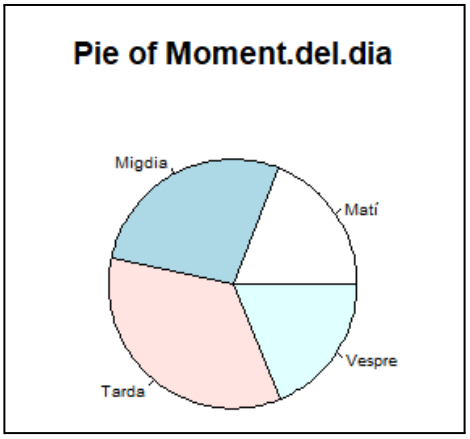
- The product line, which in the pie chart and in the bar plot we can see that it follows a uniform distribution. Even though there is quite some difference between the one that has the greatest number of products (fashion accessories) and the lowest (health and beauty).





- The time of the day is a variable that we added ourselves to the database. It is clearly seen that the majority of people purchase at the supermarket at the middle start of the day, at noon or afternoon, instead of going at the start or the end of the day.

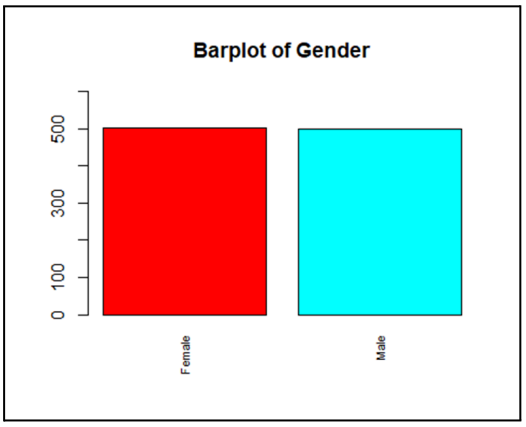


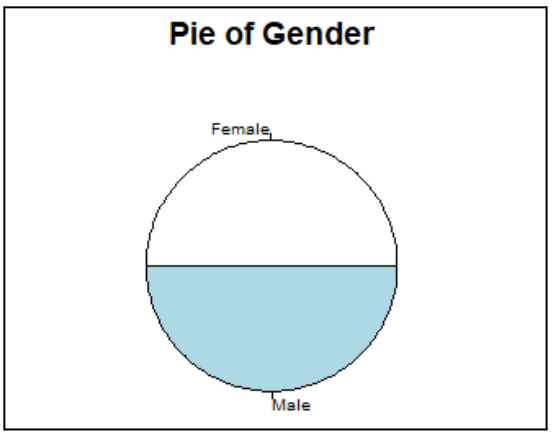


- <u>Binary variables:</u>

  There are two binary variables in our database, of which all of them have a barplot and a pie chart, are the following ones:

  - The gender of the customer, female or male. We can see that there isn't any appreciated difference between the customers that are male and the ones that are female. As that's a normal thing in real life, because the number of males and females is quite the same and in normal life the same number of each of them go shopping.





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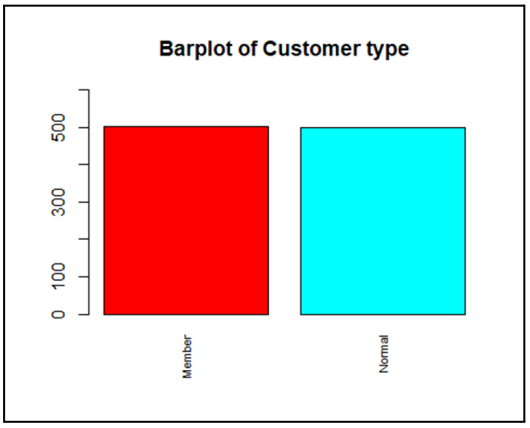
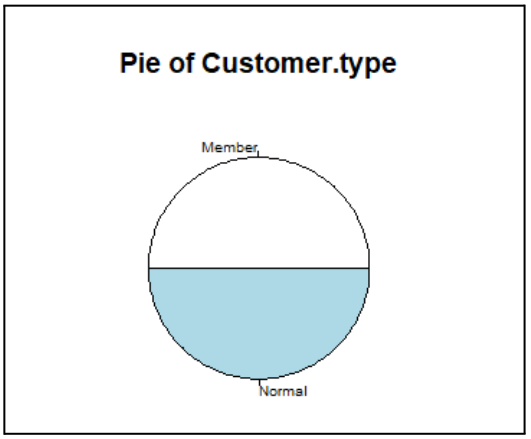
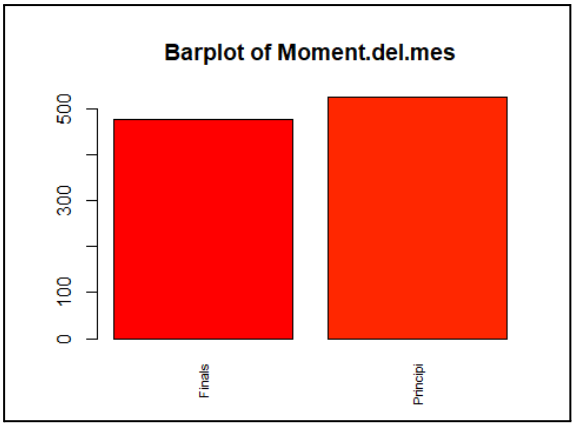
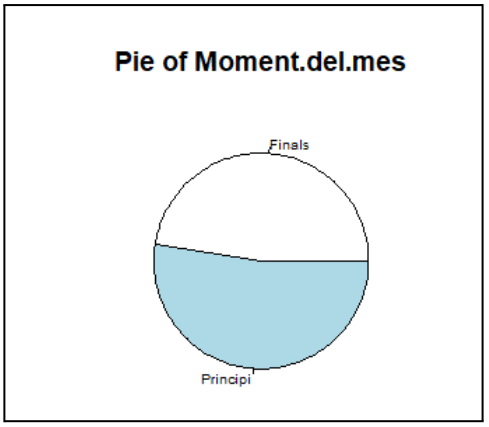
  - The type of the customer, member or normal. And it's clear that there isn't any seeable difference between the customers that are members and the ones that are not, which is quite strange that this univariable graphics follows an uniform distribution too.



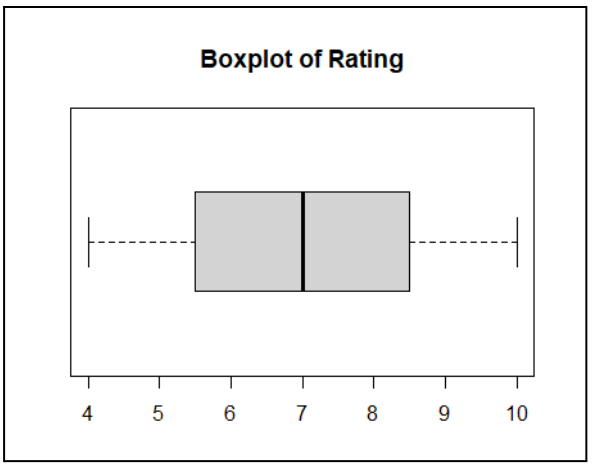


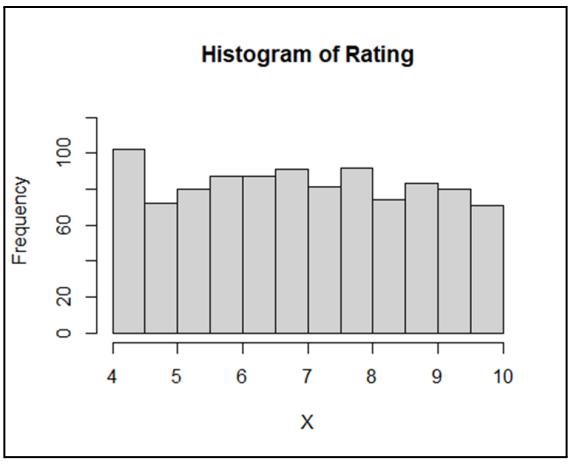
  - The time of the month is another variable that we have added ourselves to the database. This one represents if the customer has done their purchase at the start of the month or at the end of the month.





- Ordinary variable:
    - There is only one ordinary variable in our database, which is the rating of the products. In this case we have a boxplot and a histogram, whose ratings are quite diverse, as there are many that are 4 but the other numbers have some frequency too. Even though in the box plot it can be thought the distribution is more uniform than it is, it actually follows a normal distribution.



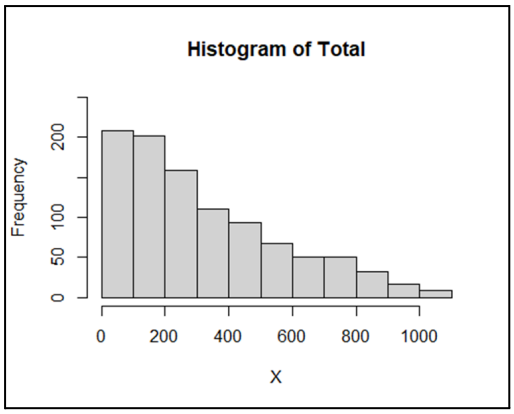


Numerical variables:

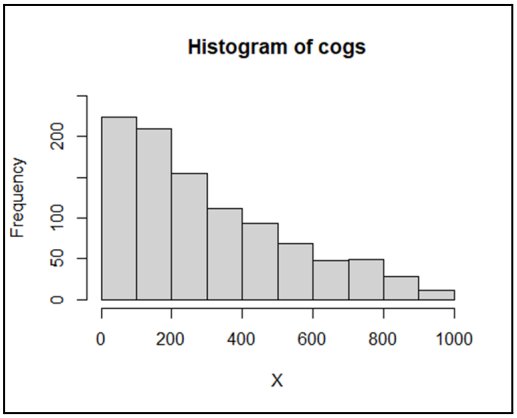
The numerical variables are divided into continuous and discrete ones.

- Continuous variables:

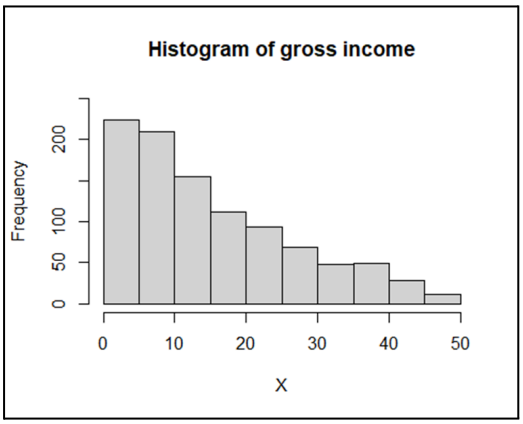
  In the continuous variables we have used histograms to represent them.



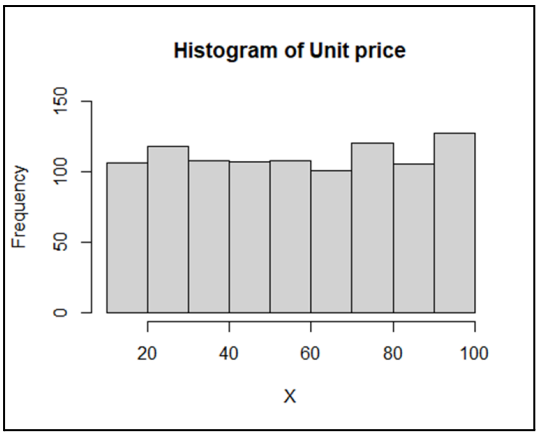
- The total money spent including tax: we can see that it follows a gamma distribution. As there are less bills that have a greater total money spent than the ones that have a smaller amount.



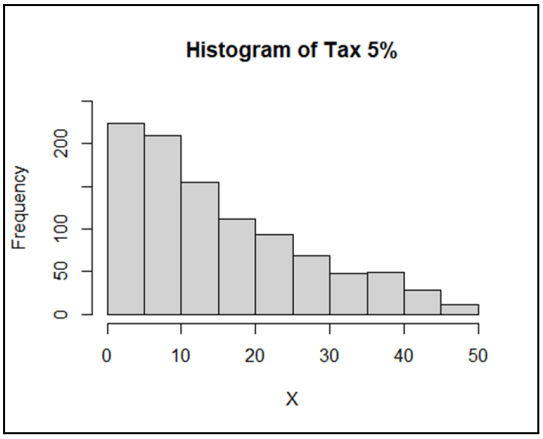
- The cost of goods sold, which is the sum of every article price. In this case, we see a gamma distribution too, as the total and the cost of each food is interconnected.



-The gross income, which is the earnings before taxes or other deductions. This histogram follows the same pattern as the cost of goods sold graph.

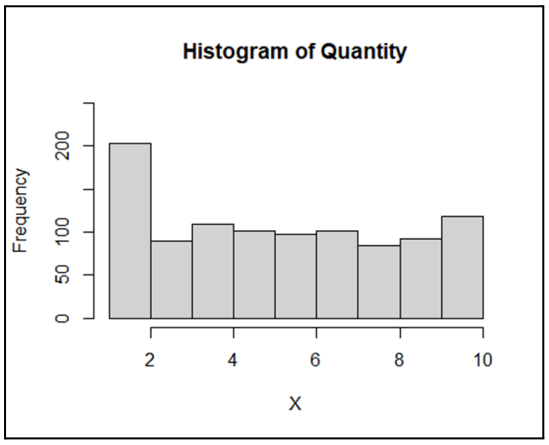


-The unit price, which is the average price for a unit, follows a uniform distribution with a tendency to a normal distribution.



-The tax price that adds to the price of each product (a 5% tax rate for each purchase). As the tax price is the 5% of the cost of each good, it follows the same pattern as that variable.

- <u>Discrete variables</u>:

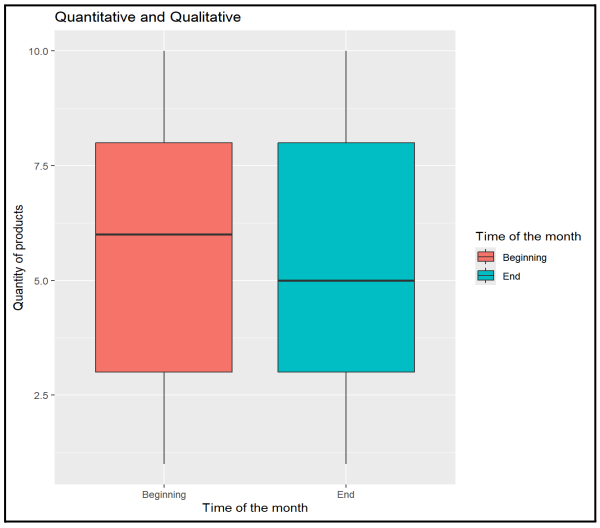


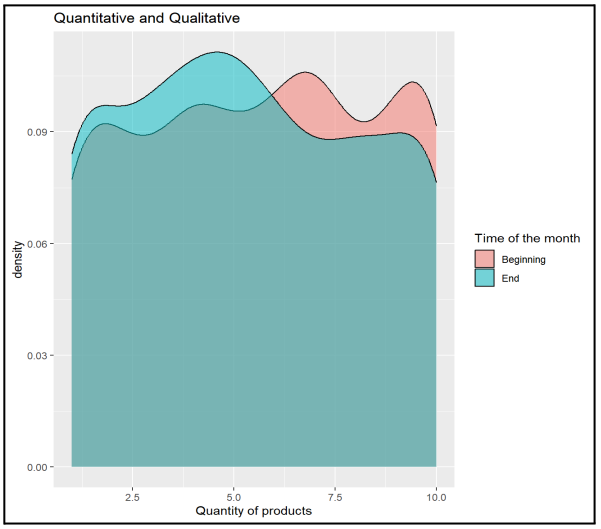
The only discrete variable is the quantity, in other words the amount of articles purchased. The graphical representation is an histogram.

b. BIVARIANT

For our Bivariate statistics we chose the following variables: Time of the month and Quantity (Qualitative and quantitative), Gender and Product Line (Two Qualitatives) and Quantity and Tax (Two Quantitatives).

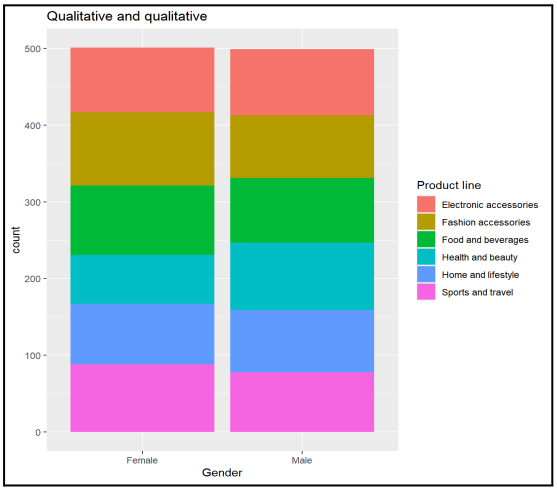
<u>Quantitative and Qualitative (Time of the month / Quantity)</u>

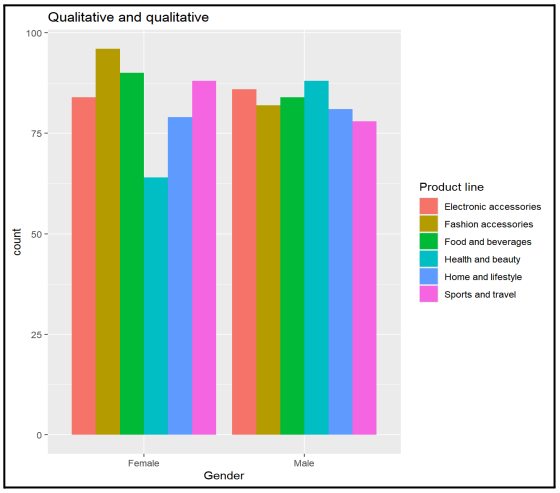




Looking at the box plots, we can see that the quantity of products bought at the beginning of the month is not much different from the quantity of products bought at the end of the month. The only significant difference we can see is the mean, which is a bit higher in the beginning (5,65) than in the end (5,36), and the median, which is also higher in the beginning. At the end of the month, we see that the most frequent quantity is 4, and the higher quantities are less frequent.

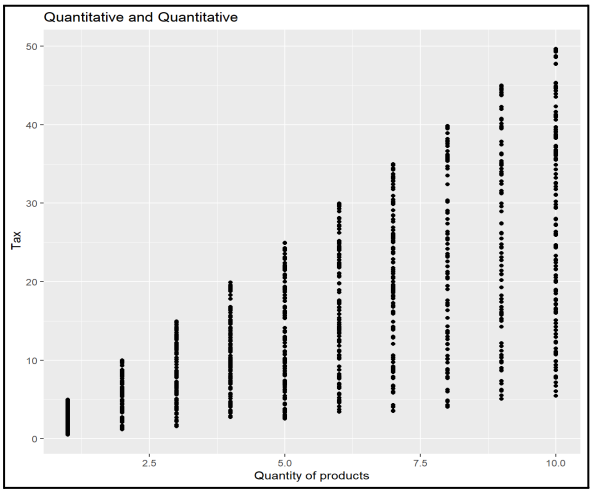
Two Qualitatives (Gender / Product Line)





The biggest difference between the products bought by men and by women is Health and Beauty products, where we have 88 observations of male customers buying health and beauty products, and 64 observations of females. Another difference is Fashion accessories, with 82 observations for men and 96 for women. We can also see that Sports and Travel observations are higher for women. For the rest, there isn't a big difference between men and women.

Two Quantitatives (Quantity / Tax)



First of all we can see that the graph forms straight lines and there aren't any dots between these lines, that is because the Quantity of products is a discrete variable and it's impossible to find an observation with a decimal number of products. We can also see that these lines form sort of a triangle shape, that is because when the quantity of products is low the price must also be low, but when there is a higher quantity, the total can be very high (if a lot of expensive things were bought) or not

that high (if a lot of cheap things were bought), and because the total and the tax are proportional the dots form a triangular shape.

c. CONCLUSIONS

Analyzing the categorical variables, which include nominal, binary, and ordinal types, various distributions were observed. Nominal (city, type of payment, product line, and time of day) and binary (customer gender and membership status) variables showed generally uniform distributions. The ordinal variable, product rating, indicated a more diverse distribution, approximating normality. Numerical variables, both continuous and discrete, revealed patterns such as a gamma distribution, uniform and normal distributions. With the univariate statistics we can conclude that all the continuous numerical variables are mutually dependent between each other, as the cost of goods, the total, taxa and gross income all depend on the total of the purchase. From the bivariate we can conclude that most of the variables don't have any relation between each other, and they don't depend on one another, except for Total, Cogs, Tax 5%, gross margin percentage, which have a very strong relationship with each other following an almost direct relationship.

6. DESCRIPTIVE ANALYSIS OF SAMPLE DATA

To sample our database we decided to use cluster random sampling. This type of sampling involves dividing the population into different random clusters of individuals. Each of these is naturally heterogenic, as for our database this was the best.

```
mean(dd$Total)
sd(dd$Total)

df <- data.frame(tour = rep(1:10, each=80),experience = rnorm(800, mean=7, sd=1))

clusters <- sample(unique(df$tour), size=80, replace=F)
```

In this code we can see how we divided our database into a set of random clusters and then from these clusters we selected the sample, which is the cluster that we are going to study.

At first we thought that it could be a better idea to use stratified random sampling, but we discovered that when our database consists of very heterogeneous cities, so the best type of sampling is the cluster one.

7. CONFIDENCE INTERVAL

The confidence interval shows a range of values derived from sample data, used to estimate an unknown population parameter. In our script the confidence level is set to 95%, so this percentage of the intervals will contain the true population parameter, while about 5% will not.

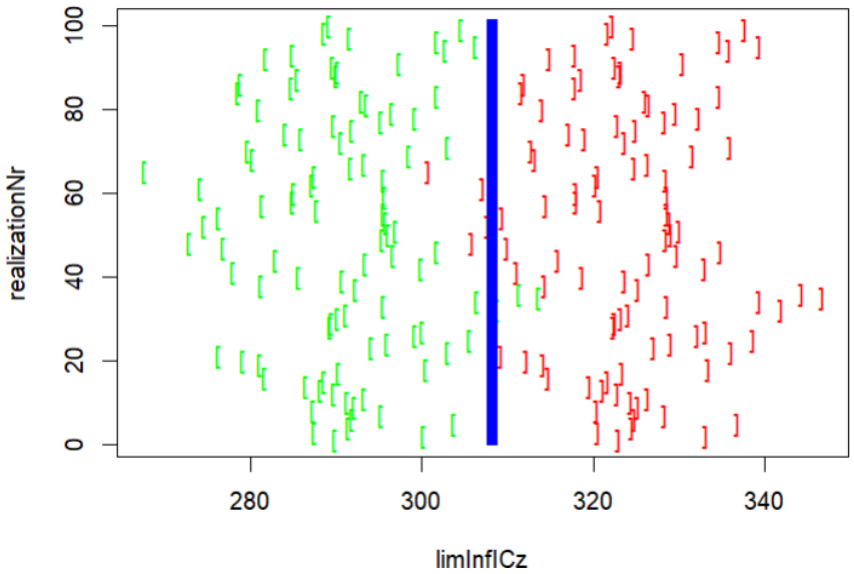
To achieve more accurate and stable estimates, we use a sample size of 800, which is 80% of the dataset, which reduces the variability leading to more precise statistical inferences. The parameter to be estimated in this analysis is the population mean of the variable "Total".

It is realistic to estimate the mean because it is a fundamental measure that provides much information to summarize the overall behavior of the dataset. Moreover, as the dataset contains a representative sample, the mean of the "Total" will likely provide a good approximation of the central value.

Otherwise, in many scenarios, understanding the average value of a variable is crucial, so choosing the Total price of products can provide us relevant information from the database while other variables like cogs or tax, which although they are directly proportional, are more confusing to analyze.

To realize this estimation we use two procedures to calculate the confidence interval: using the Z-distribution, assuming that the standard deviation is known; and the T-distribution, where the standard deviation is assumed to be unknown.

- **Z-distribution**



$$\bar{x} \pm z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

$\bar{x}$ = sample mean

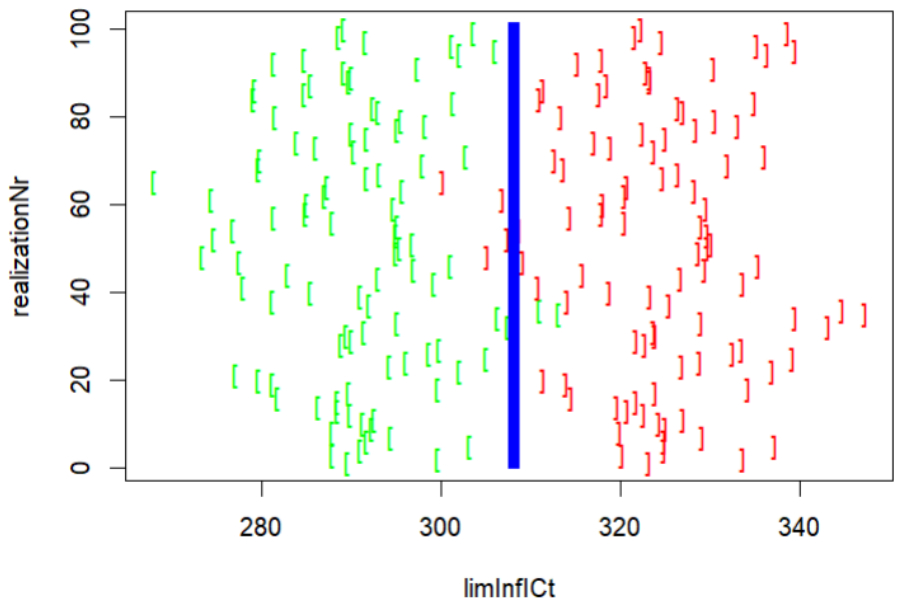
Z-score for the confidence level

σ: population standard deviation

n: sample size

7 failings

- **T-distribution**



$$\bar{x} \pm t_{\frac{\alpha}{2},n-1} \frac{\sigma}{\sqrt{n}}$$

$\bar{x}$ = sample mean

T-score with n-1 degrees of freedom for the desired confidence level

σ: sample standard deviation

n: sample size

6 failings

Both graphs show the confidence intervals for multiple sample realizations from a dataset. We can see green and red brackets indicating the lower and upper limits of intervals and a vertical blue line representing the population mean. The vertical axis represents the different sample realizations (100) and the horizontal axis the limit of the confidence intervals.

Most of the intervals (green to red brackets) include the blue line, indicating that the true mean is within the confidence interval for those realizations. If an interval does not contain the blue line, it indicates that for that particular sample realization, the confidence interval did not capture the true mean.

By visual inspection, you can see how many intervals do not cover the true mean. This is a direct visualization of the confidence level in practice. When the population standard deviation is known or the sample size is large, the Z-distribution provides more precise confidence intervals; while when the population standard deviation is unknown and the sample size is small, the T-distribution is more appropriate, as it provides wider intervals that better reflect the uncertainty.

In our case, both graphics should show a high percentage of intervals covering the true mean, around 95%, which is proven as there are only 7 missings in Z-distribution and 6 missing in T-distribution, where both values are around 5%, so our results seem to be reliable.

8. HYPOTHESIS TEST

first scenario

The store manager of a supermarket chain has been in his position for more than 10 years. He suspects that the average number of products that customers buy is greater than 5. To check his prediction, he tracks the number of products that each customer buys over a certain period of time.

hypothesis: *The average number of products that customers buy is greater than 5.*

In order to test this hypothesis, we will use the t.test, since we are dealing with a numeric variable. In this scenario the null hypothesis is that the mean number of products purchased is less than or equal to 5, the alternative is that the mean is greater than 5.

- H0: mean ≤ 5
- HA: mean > 5

```
        One Sample t-test

data:  x
t = 3.8143, df = 799, p-value = 7.356e-05
alternative hypothesis: true mean is greater than 5
95 percent confidence interval:
 5.226595      Inf
sample estimates:
mean of x
  5.39875
```

- **t-value:** A t-value of 3.8143 indicates that the sample mean is significantly above the hypothesized mean of 5.
- **p-value:** The p-value of 7.356e-05 is much smaller than the commonly used significance level of 0.05. This provides strong evidence against the null hypothesis.
- **Confidence interval:** The confidence interval for the mean does not include the value 5, which reinforces the conclusion that the true mean is greater than 5.

We reject the null hypothesis because the p-value obtained is extremely small, indicating a very low probability that the mean number of products purchased is 5 or less. Furthermore, the t-value of 3.8143 and the confidence interval ranging from 5.226595 to infinity reinforce this conclusion. Taken together, these results provide strong evidence that the true mean of products purchased is greater than 5, thus confirming the alternative hypothesis and supporting the manager's suspicion.

second scenario

An employee of a very successful supermarket chain is also studying Statistics at the same time. As a final assignment for one of her subjects, she decides to do a study on the supermarket chain where she works. This student is sure that there is a relationship between the gender of the customers and the type of products they buy.

hypothesis: *There is a relationship between the gender of the customers and the type of products they buy.*

In order to test this hypothesis, we will use the Chi-squared test, since we are dealing with two categorical variables. In this scenario the null hypothesis is that there is not a significant relationship between the gender of the customers and the type of products they buy; the alternative is that this relationship exists.

- H0: No relationship between gender of customers and type of products bought.
- HA: Relationship between gender of customers and type of products bought.

```
        Pearson's Chi-squared test

data:  x and y
X-squared = 9.5402, df = 5, p-value = 0.08936
```

- **Chi-square statistic:** A value of 9.5402 gives us a measure of the discrepancy between the observed and expected frequencies. However, this value alone is not sufficient to make a decision; we need to consider the p-value.
- **p-value:** The p-value of 0.08936 is greater than the commonly used significance level of 0.05. This indicates that there is insufficient evidence to reject the null hypothesis.

There is insufficient evidence to reject the null hypothesis that the two categorical variables (gender and type of products purchased, for example) are independent. In other words, no statistically significant relationship was found between the two variables in this analysis. This suggests that any observed differences in the frequencies of products purchased by different genders could be due to chance.

third scenario

Two roommates are talking about their expenses and a specific topic comes up in the conversation. One of the roommates believes that people tend to buy more products at the beginning of the month than at the end of the month because they have just been paid. The other says that there is no significant relationship between the two. They want to verify whose statement is true, so they do some research and find a database with data on purchases in a famous supermarket chain.

hypothesis: *There is a significant relationship between the time of the month of the purchase and the quantity of products bought in the purchase.*

In order to test this hypothesis, we will use the ANOVA test, since we are dealing with a numerical and a categorical variable. In this scenario the null hypothesis is that there is not a significant relationship between the time of the month of the purchase and the quantity of products bought in the purchase; the alternative is that this relationship exists.

- H0: No relationship between time of the month and quantity of products bought..
- HA: Relationship between time of the month and quantity of products bought.

```
Call:
   aov(formula = x ~ y)

Terms:
                     y Residuals
Sum of Squares    15.509  6970.289
Deg. of Freedom        1       798

Residual standard error: 2.955452
Estimated effects may be unbalanced
> summary(aov(x~y))
             Df Sum Sq Mean Sq F value Pr(>F)
y             1     16  15.509   1.776  0.183
Residuals   798   6970   8.735
```

- **p-value** : The p-value of 0.183 is greater than the commonly used significance level of 0.05. This indicates that there is insufficient evidence to reject the null hypothesis

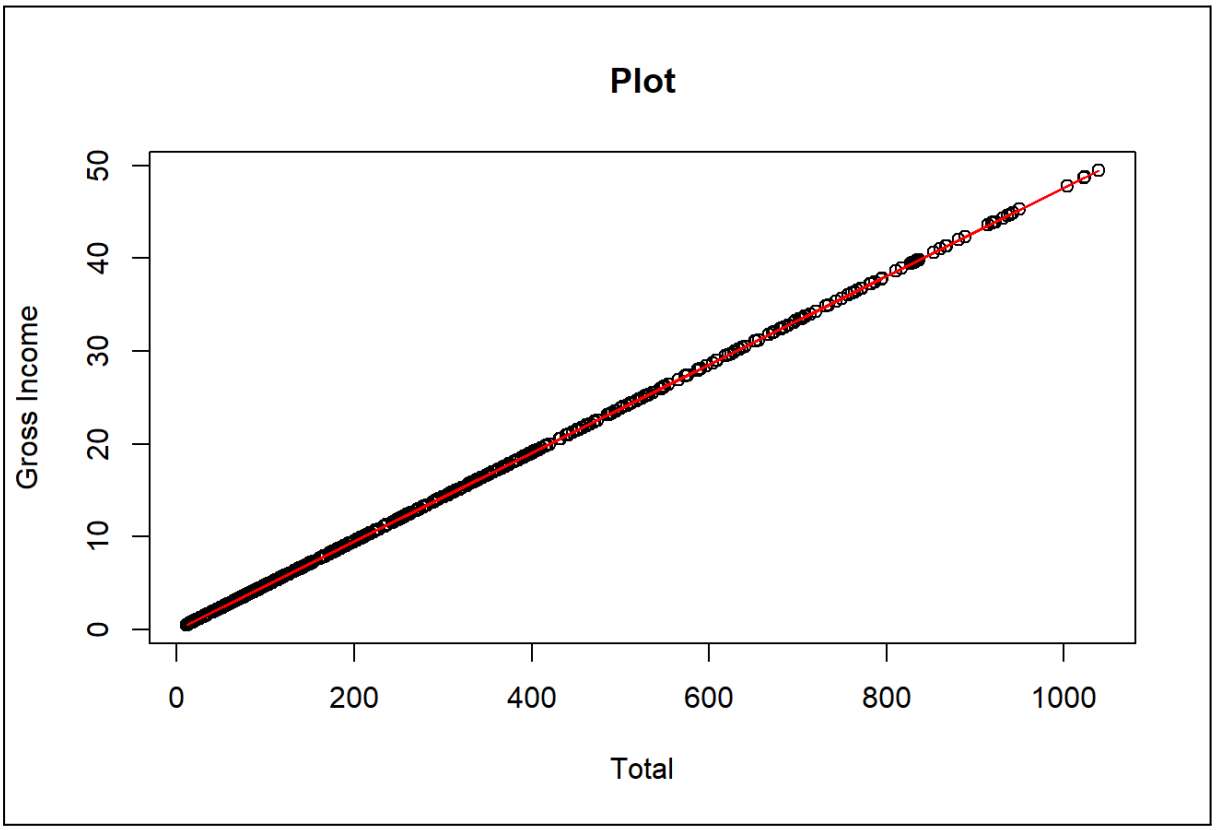
The p-value of 0.183 is greater than the commonly used significance level of 0.05. This means that there is not enough evidence to reject the null hypothesis that there is no significant difference in the quantity of products purchased as a function of the time of the month. In other words, based on these results, we cannot conclude that the time of the month has a significant effect on the quantity of products purchased.

9. REGRESSION MODEL

Our regression model obtained is a linear model (lm) used to predict 'gross income' based on the predictor variable 'Total' (gross income is regressed on Total). The formula for the model is:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Where Y is the dependent variable, 'gross income', X is the independent variable, 'Total', $\beta 0$ is the intercept, $\beta 1$ is the coefficient for X, and ϵ is the error term.



The model summary reveals highly significant coefficients. The estimated intercept is approximately -2.814e-14, which is essentially zero, indicating that when 'Total' is zero, 'gross income' is extremely close to zero. The estimated coefficient for 'Total' is 0.04762, meaning that for each unit increase in 'Total', the 'gross income' increases by approximately 0.04762 units. The standard errors of the coefficients are very small at 1.688e-15 and 1.688e-15, indicating precise estimates. The t-values for both the intercept and the Total variable are extremely high, leading to highly significant p-values (<2e-16), suggesting that both the intercept and the slope are statistically significant, this is emphasized with the significance code '***' which indicates significance.

The residuals we obtained were:

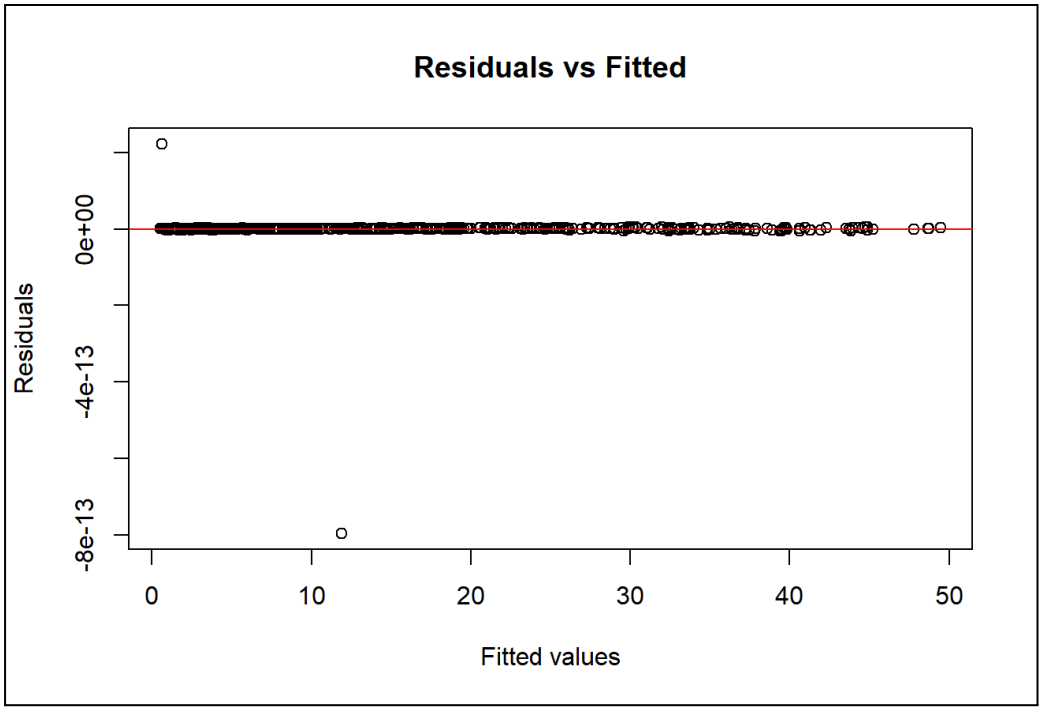
Min	1Q	Median	3Q	Max

-7.962e-13 -8.000e-17 6.900e-16 1.620e-15 2.232e-13

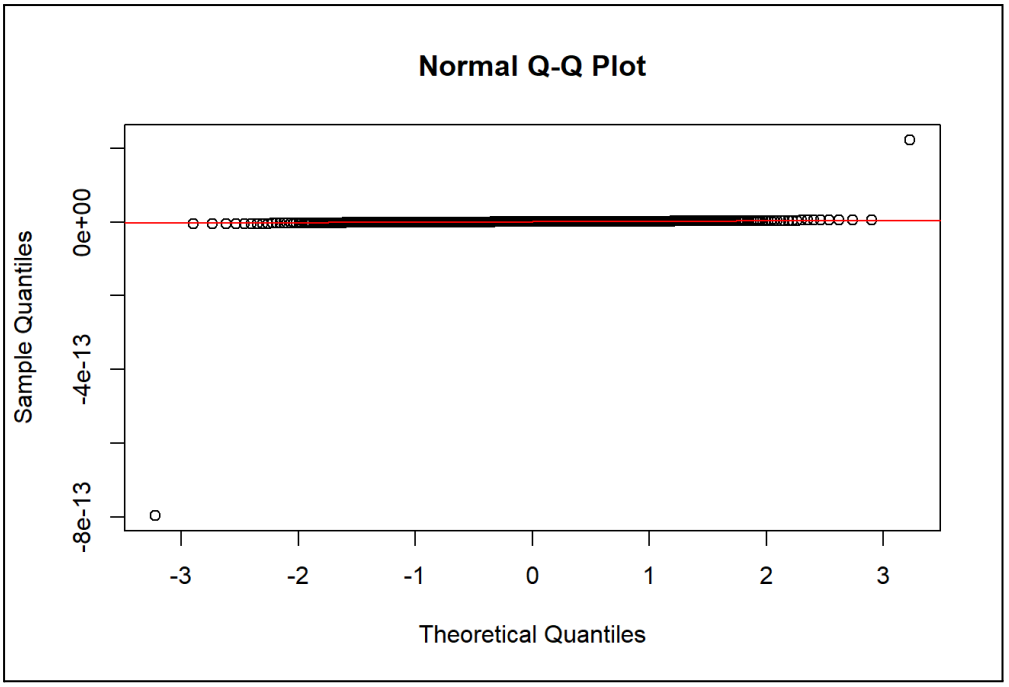
The residuals are the differences between the observed values of 'gross income' and the values predicted by the model. They are very small, suggesting that the model predictions are very close to the actual data points.

The residual standard error is extremely small, 2.932e-14, the multiple R-squared and the adjusted R-squared are equal to 1, and the F-statistic is extremely large (1.213e+32) and the corresponding p-value is very small (< 2.2e-16). These values indicate that the model is highly significant.

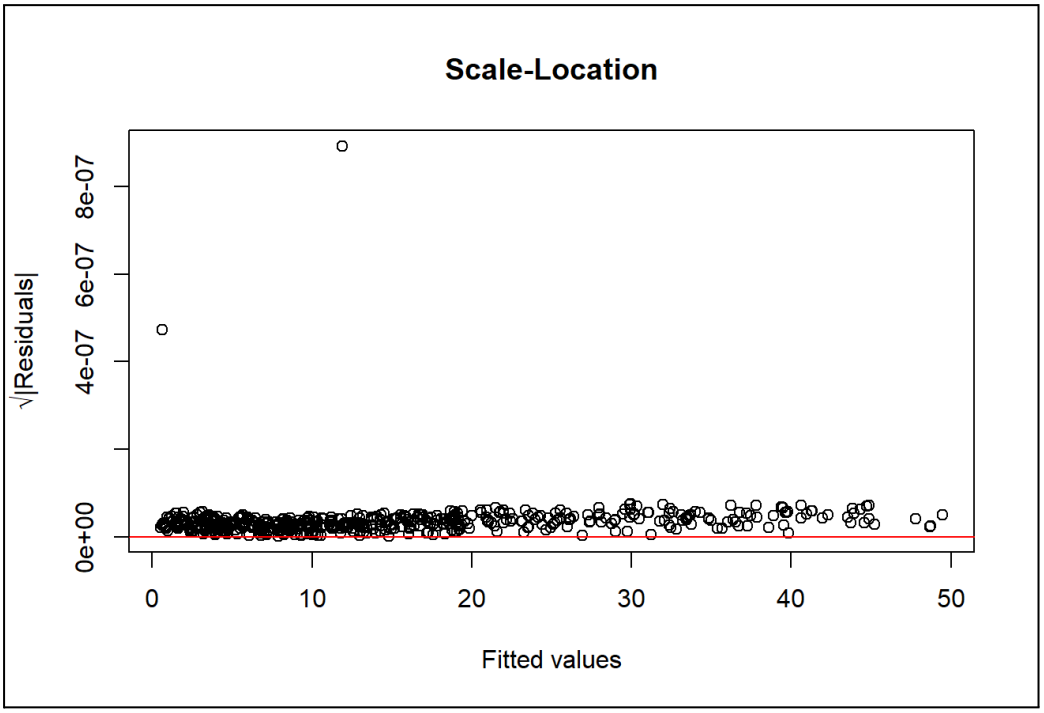
To ensure the model's validity, a graphical residual analysis was conducted. This analysis involved four key diagnostic plots: Residuals vs Fitted, Normal Q-Q Plot, Scale-Location Plot, and Residuals vs Leverage.



The Residuals vs Fitted plot shows the residuals clustered very closely around the zero line, suggesting that the model predictions are extremely accurate.

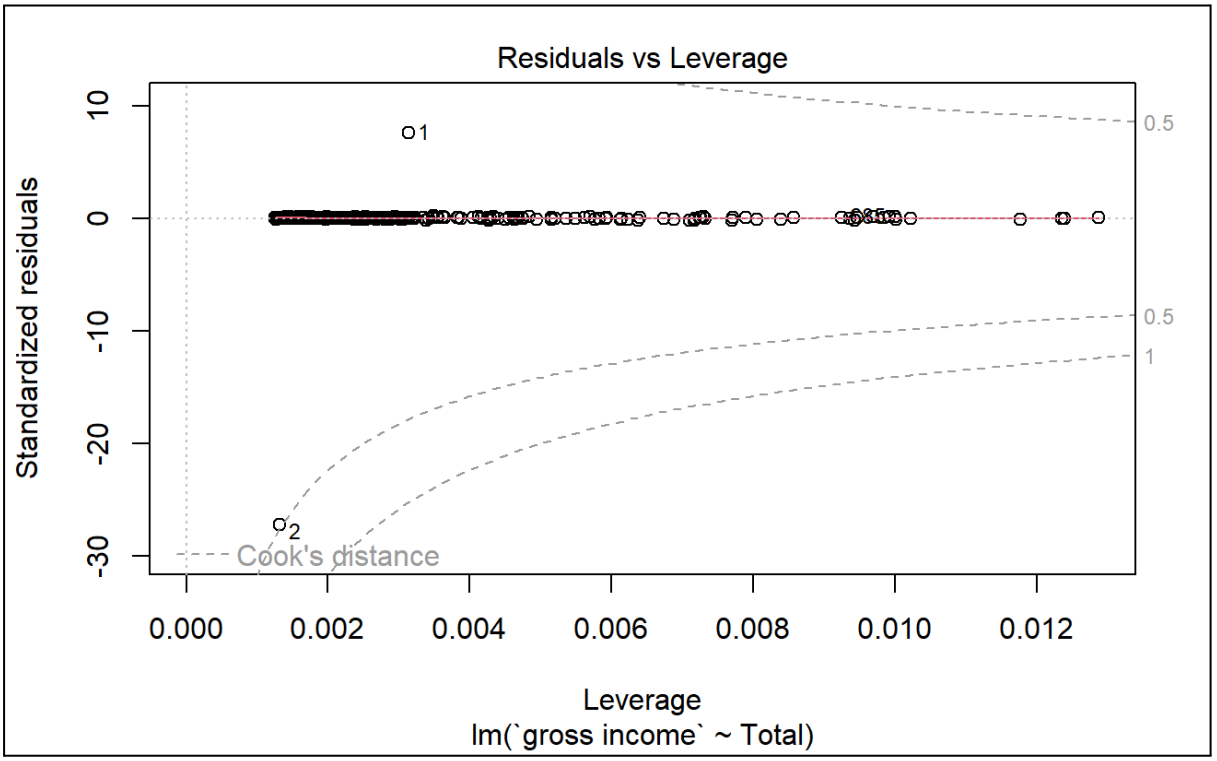


The points in the Normal Q-Q Plot lie almost perfectly along the reference line, which is a strong indication that the residuals follow a normal distribution. This conformity to the reference line suggests that the assumption of normality of residuals is met for this regression model. Given the extremely small residuals seen in the Residuals vs Fitted plot, this outcome is expected and reinforces the reliability of the model.



The Scale-Location plot is used to check the homoscedasticity, or constant variance, of residuals in a regression model. In this plot, the square root of the absolute value of the residuals is plotted against

the fitted values. We can see that the points are all very close to zero. This clustering indicates that the residuals' variance is relatively constant across all levels of the fitted values, meeting the assumption of homoscedasticity. There are a few points that stand out as potential outliers, but these are isolated instances and do not indicate a significant issue with the variance of the residuals.



Finally, the Residuals vs Leverage plot did not identify any influential data points, suggesting that no single observation disproportionately affects the model.

After getting these results we tried to see what would happen if we added more variables for our predictions, and we tried to predict ‘gross income’ from ‘Total’, ‘cogs’, ‘Tax’ and other variables, but since our results with one variable are so close to perfect the coefficients of the other variables were not being calculated, as they were reaching singularities because of these close to perfect fits, so we decided to keep our simplified model.

In conclusion, the regression model is a simple linear regression that uses ‘Total’ to predict ‘gross income’. The model's quality indicators, including a perfect R-squared value and highly significant coefficients, indicate an excellent fit. The graphical residual analysis supports the assumptions of linear regression, further validating the model.

Now having a linear regression equation that can very accurately predict the gross income, we only needed to actually prove this by comparing it to the 20% of the data that wasn't used in the cluster sample.

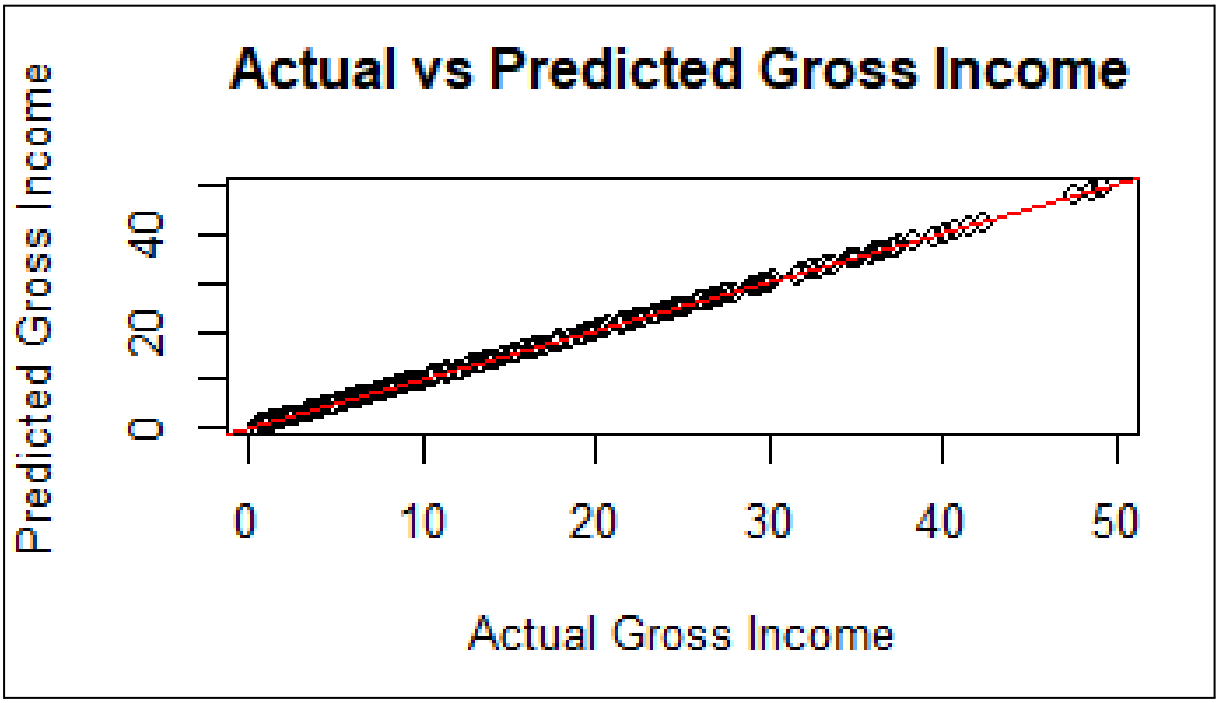
Firstly, we imported the remaining 20% of the 'gross income' data, and we calculated the difference in value between our prediction and the actual values.

```
diferencia <- sqrt((dd$`gross income`-prediccio)^2)
```

After calculating these results, we looked at its summary and we were proven to be correct about our prediction being very accurate.

```
> summary(diferencia)
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
0.000e+00 1.066e-14 1.865e-14 2.218e-14 2.842e-14 8.527e-14
```

And finally, to have a graphical representation, we made a plot using the prediction and the real values. If the predicted values are accurate, they'll form a straight line. The red line shown is a y=x line to have a reference to how close the values are to being a straight line.



As can be seen, the values form an almost perfect line, which proves that our linear regression formula is very accurate.

10. WORKING PLAN

This section of the report details the development process of our statistical analysis project, in which we have tackled a series of challenges and stages under the guidance of the team. Our group has been dedicated to meeting the established requirements and achieving the project's objectives.

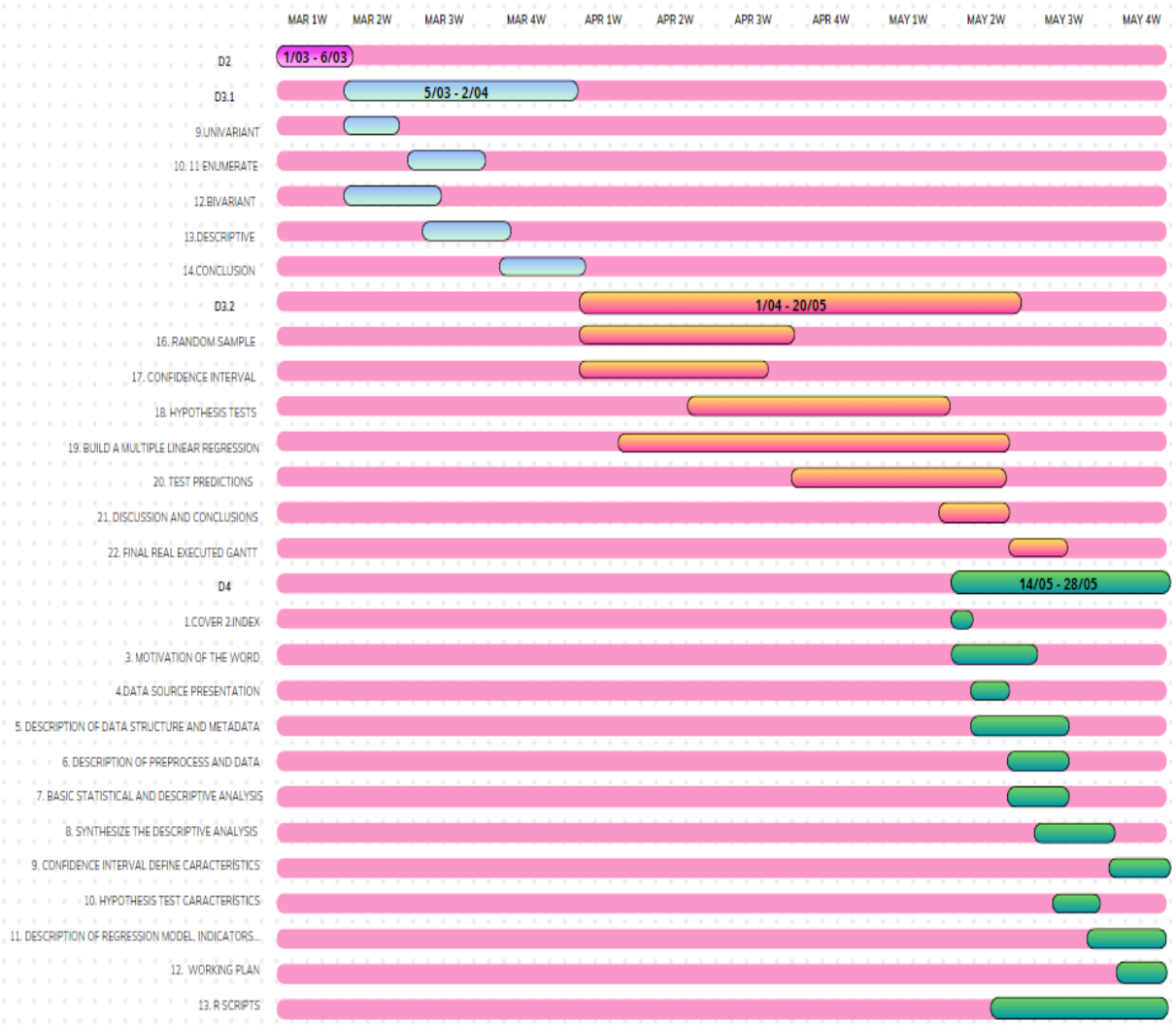
From the initial stage of defining and assigning tasks, our team has followed a detailed plan to address each phase of the work systematically and efficiently. During Phase D1, we identified the appropriate data source for our analysis, selecting a dataset related to the supermarket domain and ensuring its timely delivery. Throughout the project development, Phases D2, D3, and D4 were crucial. In Phase D2, we consolidated our team and established effective communication. In Phase D3, we carried out project development, addressing tasks such as data preparation and statistical analysis. Finally, in Phase D4, we completed and delivered the final project, highlighting the achievements and conclusions obtained throughout the process.

Throughout the process, we have faced technical and logistical challenges, as well as critical decisions regarding data structure and analysis approach. However, we have overcome these obstacles and successfully progressed with our project. Additionally, we have followed a collaborative approach and effectively divided tasks.

The first thing related to the working plan we did was the Gantt chart. This helped us organize our project and make sure we were following the guidelines that were given to us.

FIRST GANTT CHART

FINAL GANTT CHART



If we look at the final Gantt chart, we can see that although all tasks have been completed, some of them have taken us much more time than we expected. Perhaps we were too confident in ourselves, but what we know for sure is that when we did the first Gantt chart we didn't know what sampling, test predictions or multiple linear regression were and what they consisted of. Even though the time tasks took was different to what we expected, and on some occasions we were doing multiple tasks at the same time and double the work to fix unexpected issues, we have managed to stay on track and successfully complete the project through effective collaboration and determination.

Daily Reports

13th of February

Initially, our team searched for potential datasets for our statistical analysis project. The allure of exploring Pokémon data captured our collective interest. However, following a discussion with our professor, Xavi, we reconsidered our options and ultimately opted for a different database related with the supermarkets domain. This decision was reached after careful consideration of various factors, including the availability of comprehensive data, relevance to our academic objectives, and the potential for insightful analysis.

20th of February

Upon reviewing the dataset, we recognized a deficiency in qualitative variables, prompting us to think about making some new ones. One such variable we deliberated upon was time segmentation, specifically categorizing observations based on the time of day, including morning, noon, and evening.

This segmentation aimed to capture potential variations in consumer behavior and preferences across different times of the day. Additionally, we considered incorporating a variable based on the timing within the month, distinguishing between the beginning and end periods. This segmentation strategy aimed to account for potential fluctuations in consumer spending patterns or shopping habits, influenced by factors such as paydays, bill cycles, or promotional events occurring at specific intervals within the month.

By introducing these new variants we aimed to elevate the quality and utility of our dataset for subsequent statistical analyses and interpretation.

27th of February

After dedicating an entire week to create the new variables and ensuring their accuracy, we shifted our focus to generating univariate graphs for all the variables. We used the code provided in class and followed the instructions meticulously.

However, when we finally got the graphs, they just didn't look right. No matter how many times we tried, the graphs just didn't come out the way they were supposed to. We ended up spending the entire week stuck in this loop of making the graphs over and over again, hoping they'd turn out better.

We had to take a step back and rethink our approach. We went back to the code and tried to figure out where we were going wrong. Moving forward, we were determined to find a solution. We found some mistakes in our database and after checking them we finally got those graphs looking the way they should.

5th of March

We worked on the D2 report, organizing the tasks we had to do, and on top of that, we also got started on creating three bivariate graphics using the code we received.

12th of March

We made progress with the bivariate statistics. Regarding the preprocessing process, we've enumerated the specific steps utilized with our dataset.

19th of March

Theoretical-practical lesson about probability and its implementations and methods in RStudio.

EASTER HOLIDAY AND MIDTERM EXAMS

16th of April

- Data sampling: we thought about doing stratified random sampling first, but it was wrong. The correction option was cluster sampling for the cities. We had problems about using 0.8N
- Review about the partial presentation, we've had noted the errors to solve them later

23rd of April

We asked the teacher about some issues we were having with the creation of the cluster sample. Then we proceeded to take the next step and we made a plot

30th of April

We started working on the confidence interval applied to our database. At the same time, some members of the group proceeded to check the errors we had in our last presentation (extend the y-axis

in the univariate plots, change the bivariate plots where it's possible to confuse the mean with the median…)

7th of May

Some members of the group started working on the hypothesis test of our database while others tried to finish the confidence interval task, as we were having some problems with it.

14th of May

We were encouraged to do a regression model applied to our database. We had many problems with it due to the relation between our variables. Some of our variables are totally related to each other, whereas others are not related at all. We talked about it with our teacher and discussed what should be done.

21th of May

As we were having many problems with our regression model we asked for our teachers' help. We finished working on the regression model and reviewed the hypothesis test and confidence interval.

11. CONCLUSIONS

Having finished this project, we can come to many conclusions. For example, the interpretations that can be made from each univariate or bivariate graph, some of the most important are: The leading number of people shop in the evening and buy only 2 or 3 products, the customers that buy from the supermarket are half male and half female, the majority of people who shop for 9 or more products do so at the beginning of the month, and the fact that women buy a notable amount more of fashion accessories and men buy significantly more health and beauty products.

However, the more important conclusions to be taken from this project have to do with the database in general.

Firstly, about the confidence interval, our visualizations reveal that while both methods aim for 95% coverage of the true mean, where there are only 7 missings in Z-distribution and 6 missing in T-distribution confidence intervals, so our results seem to be reliable as the values are around 5%.

Second, what we have concluded from the hypothesis tests is that even though it looked like some variables were related, such as gender and product line, it's important to get objective evidence such as the tests to prove whether they are related or not.

Lastly, and most importantly, is about the regression model. Making this linear regression model made us be aware about how related our variables are with one another, since the function ended up having barely any error. Which meant that our prediction ended up being very accurate.

To sum it all up, although we've had several problems throughout the time of this project, we've found out what the issues were, how to solve them and extract the desirable results that we wanted, and create this final report that we believe is satisfactory.